SLEEVEDRIFT · MARKET AND VIABILITY

SleeveDrift market and viability

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue at the kit 1.9.1 upgrade; prices cited with the date checked, unknowns marked to assess

SleeveDrift market and viability

The users are disaster-response forces and tunnel contractors that may have to dig by hand in a rescue pipe when a machine stops; they buy few kits, rarely, and spend on readiness rather than on volume, so a kit that costs about USD 10,800 in parts is a standby item, not a consumer product. Open design matters because no public design for this kit exists (SVD-PRB-001, Prior work), so forces and workshops in countries with tunnelling and collapse risk can build from the files.

> **Safety:** SleeveDrift is published as an open engineering reference, not certified rescue equipment. A force that holds a built kit trains with it and decides when to use it (SVD-REG-001).

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
National and state disaster-response forces	NDRF and state forces, India first (SVD-DDR-001 item 11); other forces in tunnelling countries	Nothing sold for pipe hand-mining. A vertical confined-space tripod with winch and 3-way SRL, the Frontline Megapod, lists at USD 4,245.35 (De Gemmill, checked 2026-11-02); it is a different function (top entry), not a substitute	10,804 in parts for option A (bom/bom.csv); labour, transport, import and training to assess	A documented hood, haul and air kit that can be staged and built locally from stock parts
Tunnel contractors	Road, rail and hydropower tunnel projects in India, Europe and elsewhere	The cost of a stopped machine and a standby rescue plan is the real comparator; the price of an auger-boring rescue spread to assess	As above	Equipment held on site as part of emergency planning; the design is theirs to adapt
Mine rescue and technical-rescue teams	Mine rescue stations, fire-service technical rescue	Four-gas monitors cost GBP 776.52 excluding VAT for the Honeywell BW GasAlertMicroClip XL (Hazchem Safety, checked 2026-11-02); the kit's monitor line (BOM 25) is USD 550 each	Gas monitors 1,100 of the 10,804	Parts that fit a rescue kit they already know (monitors, winches, rope)
Fabrication workshops and universities	Steel workshops near tunnel projects; engineering departments	Design files for rescue kits: none found for this function (to assess in a second search)	Made parts USD 1,637 of the 10,804, from the BOM (the lines marked make: 1 to 3, 5, 7 to 10, 15, 16, 27, 33)	Make the kit or teach from it; the build plan and drawings are open

Open route

Who would build it: a general steel workshop with plate rolls, near a tunnel programme or a rescue force's store (PR-014). How it spreads: through the co-design partner's own trials and training, through engineering departments that adopt it for teaching, and through rescue-equipment networks. What keeps it open: CERN-OHL-S-2.0 for the hardware, MIT for the scripts, and the safety notes in every document, so a copy carries the warnings with it.

Viability

- **Demand:** one real incident (Silkyara, 2023) shows the need; the number of forces that would buy a kit is unknown (to assess with the partner).
- **Cost:** option A is about twice the value-engineering target (PR-013). The largest lines are the winches (USD 2,300), monitors (USD 1,100), signal line (USD 860), duct (USD 765) and blower (USD 750). No price has been checked against a second supplier; every unit cost in the BOM is a first-pass estimate to confirm when parts are bought.
- **Competition:** none found for the function; adjacent products are vertical retrieval systems and gas monitors.
- **Revenue:** the project does not sell kits. Any revenue (training, build services) is for builders and partners and is outside this repo.

Sources

- De Gemmill, "Frontline Megapod Confined Space 7 ft Aluminum Tripod with 60 ft Winch and 3-Way SRL", https://degemmill.com/products/frontline-megapod-confined-space-7-ft-aluminum-tripod-with-60-ft-winch-and-3-way-srl, USD 4,245.35, accessed 2026-11-02.
- Hazchem Safety, "BW Gas Alert Micro Clip XL 4 Gas Detector (O2, LEL, CO, H2S)", https://www.hazchemsafety.com/product/bw-gas-alert-micro-clip-xl-4-gas-detector-o2-lel-co-h2s/, GBP 776.52 excluding VAT, accessed 2026-11-02.
- Silkyara reports and prior work: see SVD-PRB-001.